CYBERSRS

Campus Wireless Network Access-Control Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Build a campus wireless network access-control requirements specification. Students and guests may access the internet but must not access the internal staff network. Network administrators may manually isolate a device and restore its access. Security operators may review and acknowledge alerts but may not change access policies. The system must record administrator policy changes and retain audit records for 30 days. The planning population is 2000 registered devices; concurrency is not yet decided. Do not automatically change firewall rules or isolate devices. Requirements evolve frequently and stakeholders provide weekly stakeholder feedback. Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder project context

Detail: Build a campus wireless network access-control requirements specification.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: Students and guests may access the internet but must not access the internal staff network.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Network administrators may manually isolate a device and restore its access.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: Security operators may review and acknowledge alerts but may not change access policies.

Status: confirmed

Evidence ids: E-004

Stakeholder source statement E-005

Detail: The system must record administrator policy changes and retain audit records for 30 days.

Status: confirmed

Evidence ids: E-005

Unresolved stakeholder decision

Detail: The planning population is 2000 registered devices; concurrency is not yet decided.

Status: unresolved

Evidence ids: E-006

Out of scope: stakeholder exclusion

Detail: Do not automatically change firewall rules or isolate devices.

Status: confirmed

Evidence ids: E-007

Stakeholder project context

Detail: Requirements evolve frequently and stakeholders provide weekly stakeholder feedback.

Status: confirmed

Evidence ids: E-008

Unresolved stakeholder decision

Detail: Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Status: unresolved

Evidence ids: E-009

Open issues: What is the defined concurrency level for device access?; How will frequent requirement changes be tracked and managed in the development lifecycle?; Which specific regulatory frameworks may apply to this campus network?; What is the acceptable response time for policy change detection or alert review?; How will device isolation actions be validated before restoration?; The planning population is 2000 registered devices; concurrency is not yet decided.; Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - Students and guests may access the internet but must not access the internal staff network.

Statement: Students and guests may access the internet but must not access the internal staff network.

Rationale: Implements stakeholder source E-002.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A student device connected to the public network

When: The student attempts to access the internet

Then: The request is granted and connectivity is established

Method: test

Observable evidence: Test capture of successful internet access response

Given: A guest device connected to the public network

When: The guest attempts to access the internal staff network

Then: The request is denied and no connection is established

Method: test

Observable evidence: Test capture of firewall denial packet for internal staff network

Unresolved decisions: Whether 'access' implies only direct connectivity or includes proxy-based routes

FR-002 - Network administrators may manually isolate a device and restore its access.

Statement: Network administrators may manually isolate a device and restore its access.

Rationale: Implements stakeholder source E-003.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: An administrator with valid credentials and a device under management

When: The administrator initiates manual isolation of the device via console interface

Then: The device is isolated and status changes to 'isolated' in system state

Method: test

Observable evidence: Test capture of device isolation command execution and resulting state update

Given: An administrator with valid credentials and a device under management

When: The administrator initiates manual restoration of access to the device

Then: Access is restored and device status changes to 'active'

Method: test

Observable evidence: Test capture of access restoration command execution and resulting state update

Unresolved decisions: Whether restoration must be confirmed by a second operator or audit log

FR-003 - Do not automatically change firewall rules or isolate devices.

Statement: Do not automatically change firewall rules or isolate devices.

Rationale: Implements stakeholder source E-007.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-007

Acceptance criteria

Given: A test device in active network state with no prior isolation

When: The system attempts to automatically isolate the device via scheduled rule update or trigger event

Then: No isolation occurs and the device remains in active state

Method: test

Observable evidence: Test capture of firewall configuration before and after; no change detected

Given: A test device with active connection and configured rules

When: The system attempts to automatically modify firewall rules to restrict access

Then: No rule modification occurs and the existing rules remain unchanged

Method: test

Observable evidence: Test capture of firewall configuration before and after; no change detected

Unresolved decisions: Whether automatic actions are triggered by specific events (e.g., traffic anomalies) or time-based schedules

Related requirement ids: SEC-001

3. Non-functional requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

4. Security, privacy and compliance

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

SEC-001 - Security operators may review and acknowledge alerts but may not change access policies.

Statement: Security operators may review and acknowledge alerts but may not change access policies.

Rationale: Implements stakeholder source E-004.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: An operator with security privileges and an active alert in the system

When: The operator acknowledges the alert

Then: The system returns a successful acknowledgement response indicating the alert is acknowledged

Method: test

Observable evidence: Test capture of the acknowledgment response from the system

Given: An operator with security privileges and an active access policy rule

When: The operator attempts to modify the access policy

Then: The system denies the modification request and returns a permission denied error

Method: inspection

Observable evidence: Test capture of the denial response when attempting to modify policies

Given: The operator receives an alert for access policy change

When: The operator initiates the review action

Then: The system confirms the alert is acknowledged and no access policy modification occurs

Method: test

Observable evidence: Test capture of resulting state shows the alert status updated to ' acknowledged' without any change to access policies, as verified by a real-time policy integrity check

Unresolved decisions: Definition of 'acknowledged' state in alert lifecycle

SEC-002 - The system must record administrator policy changes and retain audit records for 30 days.

Statement: The system must record administrator policy changes and retain audit records for 30 days.

Rationale: Implements stakeholder source E-005.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-005

Acceptance criteria

Given: An administrator performs a policy change operation

When: The system logs the policy change event

Then: The audit record is stored in the system and remains accessible for 30 days after creation

Method: demonstration

Observable evidence: Test capture of audit records retrieved after 30 days from change timestamp

Given: The system has recorded an administrator policy change

When: The system attempts to return the record

Then: The system retains audit records for 30 days

Method: analysis

Observable evidence: Test capture of resulting state shows audit records retained for 30 days in the log repository with no data loss or corruption over time, as verified by a compliance inspection report from E-005_test_

Unresolved decisions: Exact retention period boundary (e.g., calendar day vs. system time); The source gives a retention duration; minimum/maximum interpretation and deletion timing remain unconfirmed.

Related requirement ids: FR-001

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: SEC-002

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003

7. Compliance and policy applicability matrix

Open decisions: What is the defined concurrency level for device access?; How will frequent requirement changes be tracked and managed in the development lifecycle?; Which specific regulatory frameworks may apply to this campus network?; What is the acceptable response time for policy change detection or alert review?; How will device isolation actions be validated before restoration?

8. Architecture

Summary: Proposed logical components and access-decision paths for source-stated boundaries; unresolved implementation choices remain open.

Items

Actor: Students and guests

Detail: Proposed logical endpoint for the source-stated actor: Students and guests.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Component name: Actor: Students and guests

Responsibility: Represent the source-stated actor: Students and guests.

Resource: internet

Detail: Proposed logical endpoint for the source-stated resource: internet.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Component name: Resource: internet

Responsibility: Represent the source-stated resource: internet.

Resource: internal staff network

Detail: Proposed logical endpoint for the source-stated resource: internal staff network.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Component name: Resource: internal staff network

Responsibility: Represent the source-stated resource: internal staff network.

Actor: Network administrators

Detail: Proposed logical endpoint for the source-stated actor: Network administrators.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-002

Component name: Actor: Network administrators

Responsibility: Represent the source-stated actor: Network administrators.

Actor: Security operators

Detail: Proposed logical endpoint for the source-stated actor: Security operators.

Status: proposed

Evidence ids: E-004

Requirement ids: SEC-001

Component name: Actor: Security operators

Responsibility: Represent the source-stated actor: Security operators.

Actor: Administrator

Detail: Proposed logical endpoint for the source-stated actor: Administrator.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

Component name: Actor: Administrator

Responsibility: Represent the source-stated actor: Administrator.

Access Control Engine

Detail: Proposed logical owner for FR-001, SEC-001.

Status: proposed

Evidence ids: E-002; E-004

Requirement ids: FR-001; SEC-001

Component name: Access Control Engine

Responsibility: Students and guests may access the internet but must not access the internal staff network. Security operators may review and acknowledge alerts but may not change access policies.

Manual Isolation Module

Detail: Proposed logical owner for FR-002, FR-003.

Status: proposed

Evidence ids: E-003; E-007

Requirement ids: FR-002; FR-003

Component name: Manual Isolation Module

Responsibility: Network administrators may manually isolate a device and restore its access. Do not automatically change firewall rules or isolate devices.

Audit Logging Service

Detail: Proposed logical owner for SEC-002.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

Component name: Audit Logging Service

Responsibility: The system must record administrator policy changes and retain audit records for 30 days.

Open issues: The planning population is 2000 registered devices; concurrency is not yet decided.; Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

From component: Actor: Students and guests

To component: Access Control Engine

Data or action: Students and guests may access the internet but must not access the internal staff network.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-002

From component: Actor: Network administrators

To component: Manual Isolation Module

Data or action: Network administrators may manually isolate a device and restore its access.

Proposed source-defined interaction

Detail: Proposed directed path for SEC-001.

Status: proposed

Evidence ids: E-004

Requirement ids: SEC-001

From component: Actor: Security operators

To component: Access Control Engine

Data or action: Security operators may review and acknowledge alerts but may not change access policies.

Proposed source-defined interaction

Detail: Proposed directed path for SEC-002.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

From component: Actor: Administrator

To component: Audit Logging Service

Data or action: The system must record administrator policy changes and retain audit records for 30 days.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

From component: Access Control Engine

To component: Resource: internet

Data or action: Permitted access decision. Students and guests may access the internet but must not access the internal staff network.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

From component: Access Control Engine

To component: Resource: internal staff network

Data or action: Denied access attempt. Students and guests may access the internet but must not access the internal staff network.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Untrusted component: Actor: Students and guests

Trusted component: Access Control Engine

Enforcement: Students and guests may access the internet but must not access the internal staff network.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-002

Untrusted component: Actor: Network administrators

Trusted component: Manual Isolation Module

Enforcement: Network administrators may manually isolate a device and restore its access.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for SEC-001.

Status: proposed

Evidence ids: E-004

Requirement ids: SEC-001

Untrusted component: Actor: Security operators

Trusted component: Access Control Engine

Enforcement: Security operators may review and acknowledge alerts but may not change access policies.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for SEC-002.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

Untrusted component: Actor: Administrator

Trusted component: Audit Logging Service

Enforcement: The system must record administrator policy changes and retain audit records for 30 days.

9. Threats

Threats

Id: THR-001

Asset: Source-defined permissions and restricted resources

Actor: Unconfirmed actor or cause (planning hypothesis)

Scenario: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Students and guests may access the internet but must not access the internal staff network.

Impact: Unauthorized access or modification outside the stated permission boundary could occur.

Control requirement ids: FR-001

Evidence ids: E-002

Residual risk: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

Id: THR-002

Asset: Source-defined permissions and restricted resources

Actor: Unconfirmed actor or cause (planning hypothesis)

Scenario: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Security operators may review and acknowledge alerts but may not change access policies.

Impact: Unauthorized access or modification outside the stated permission boundary could occur.

Control requirement ids: SEC-001

Evidence ids: E-004

Residual risk: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

Id: THR-003

Asset: Source-protected network state

Actor: Unconfirmed actor or cause (planning hypothesis)

Scenario: An implementation defect or misuse triggers an excluded automatic action, contrary to the explicit source boundary: Do not automatically change firewall rules or isolate devices.

Impact: Network state could change contrary to the stakeholder's explicit exclusion.

Control requirement ids: FR-003

Evidence ids: E-007

Residual risk: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

Risks

Id: RISK-001

Description: Implementation may fail to enforce the boundary concerning Source-defined permissions and restricted resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Students and guests may access the internet but must not access the internal staff network.

Mitigation: Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-001

Evidence ids: E-002

Owner: TBD

Id: RISK-002

Description: Implementation may fail to enforce the boundary concerning Source-defined permissions and restricted resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Security operators may review and acknowledge alerts but may not change access policies.

Mitigation: Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: SEC-001

Evidence ids: E-004

Owner: TBD

Id: RISK-003

Description: Implementation may fail to enforce the boundary concerning Source-protected network state.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: An implementation defect or misuse triggers an excluded automatic action, contrary to the explicit source boundary: Do not automatically change firewall rules or isolate devices.

Mitigation: Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-003

Evidence ids: E-007

Owner: TBD

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: Students and guests

Goal: Students and guests may access the internet but must not access the internal staff network.

Story: As Students and guests, I need the following behavior: Students and guests may access the internet but must not access the internal staff network.

Preconditions

1. A student device connected to the public network

Steps

1. The student attempts to access the internet

Success outcome: The request is granted and connectivity is established

Requirement ids: FR-001

Evidence ids: E-002

Unresolved decisions: Whether 'access' implies only direct connectivity or includes proxy-based routes

Id: UC-002

Actor: Students and guests

Goal: Students and guests may access the internet but must not access the internal staff network.

Story: As Students and guests, I need the following behavior: Students and guests may access the internet but must not access the internal staff network.

Preconditions

1. A guest device connected to the public network

Steps

1. The guest attempts to access the internal staff network

Success outcome: The request is denied and no connection is established

Requirement ids: FR-001

Evidence ids: E-002

Unresolved decisions: Whether 'access' implies only direct connectivity or includes proxy-based routes

Id: UC-003

Actor: Network administrators

Goal: Network administrators may manually isolate a device and restore its access.

Story: As Network administrators, I need the following behavior: Network administrators may manually isolate a device and restore its access.

Preconditions

1. An administrator with valid credentials and a device under management

Steps

1. The administrator initiates manual isolation of the device via console interface

Success outcome: The device is isolated and status changes to 'isolated' in system state

Requirement ids: FR-002

Evidence ids: E-003

Unresolved decisions: Whether restoration must be confirmed by a second operator or audit log

Id: UC-004

Actor: Network administrators

Goal: Network administrators may manually isolate a device and restore its access.

Story: As Network administrators, I need the following behavior: Network administrators may manually isolate a device and restore its access.

Preconditions

1. An administrator with valid credentials and a device under management

Steps

1. The administrator initiates manual restoration of access to the device

Success outcome: Access is restored and device status changes to 'active'

Requirement ids: FR-002

Evidence ids: E-003

Unresolved decisions: Whether restoration must be confirmed by a second operator or audit log

Id: UC-005

Actor: Security operators

Goal: Security operators may review and acknowledge alerts but may not change access policies.

Story: As Security operators, I need the following behavior: Security operators may review and acknowledge alerts but may not change access policies.

Preconditions

1. An operator with security privileges and an active alert in the system

Steps

1. The operator acknowledges the alert

Success outcome: The system returns a successful acknowledgement response indicating the alert is acknowledged

Requirement ids: SEC-001

Evidence ids: E-004

Unresolved decisions: Definition of 'acknowledged' state in alert lifecycle

Id: UC-006

Actor: Security operators

Goal: Security operators may review and acknowledge alerts but may not change access policies.

Story: As Security operators, I need the following behavior: Security operators may review and acknowledge alerts but may not change access policies.

Preconditions

1. An operator with security privileges and an active access policy rule

Steps

1. The operator attempts to modify the access policy

Success outcome: The system denies the modification request and returns a permission denied error

Requirement ids: SEC-001

Evidence ids: E-004

Unresolved decisions: Definition of 'acknowledged' state in alert lifecycle

Id: UC-007

Actor: Security operators

Goal: Security operators may review and acknowledge alerts but may not change access policies.

Story: As Security operators, I need the following behavior: Security operators may review and acknowledge alerts but may not change access policies.

Preconditions

1. The operator receives an alert for access policy change

Steps

1. The operator initiates the review action

Success outcome: The system confirms the alert is acknowledged and no access policy modification occurs

Requirement ids: SEC-001

Evidence ids: E-004

Unresolved decisions: Definition of 'acknowledged' state in alert lifecycle

Open issues: Whether 'access' implies only direct connectivity or includes proxy-based routes; Whether restoration must be confirmed by a second operator or audit log; Definition of ' acknowledged' state in alert lifecycle

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A student device connected to the public network
2. When: The student attempts to access the internet

Expected result: Expected: The request is granted and connectivity is established

Evidence to record: Evidence: Test capture of successful internet access response

Test type: positive

Id: TEST-002

Requirement ids: FR-001

Procedure

1. Given: A guest device connected to the public network
2. When: The guest attempts to access the internal staff network

Expected result: Expected: The request is denied and no connection is established

Evidence to record: Evidence: Test capture of firewall denial packet for internal staff network

Test type: positive

Id: TEST-003

Requirement ids: FR-002

Procedure

1. Given: An administrator with valid credentials and a device under management
2. When: The administrator initiates manual isolation of the device via console interface

Expected result: Expected: The device is isolated and status changes to 'isolated' in system state

Evidence to record: Evidence: Test capture of device isolation command execution and resulting state update

Test type: positive

Id: TEST-004

Requirement ids: FR-002

Procedure

1. Given: An administrator with valid credentials and a device under management
2. When: The administrator initiates manual restoration of access to the device

Expected result: Expected: Access is restored and device status changes to 'active'

Evidence to record: Evidence: Test capture of access restoration command execution and resulting state update

Test type: positive

Id: TEST-005

Requirement ids: FR-003

Procedure

1. Given: A test device in active network state with no prior isolation
2. When: The system attempts to automatically isolate the device via scheduled rule update or trigger event

Expected result: Expected: No isolation occurs and the device remains in active state

Evidence to record: Evidence: Test capture of firewall configuration before and after; no change detected

Test type: negative

Id: TEST-006

Requirement ids: FR-003

Procedure

1. Given: A test device with active connection and configured rules
2. When: The system attempts to automatically modify firewall rules to restrict access

Expected result: Expected: No rule modification occurs and the existing rules remain unchanged

Evidence to record: Evidence: Test capture of firewall configuration before and after; no change detected

Test type: negative

Id: TEST-007

Requirement ids: SEC-001

Procedure

1. Given: An operator with security privileges and an active alert in the system
2. When: The operator acknowledges the alert

Expected result: Expected: The system returns a successful acknowledgement response indicating the alert is acknowledged

Evidence to record: Evidence: Test capture of the acknowledgment response from the system

Test type: positive

Id: TEST-008

Requirement ids: SEC-001

Procedure

1. Given: An operator with security privileges and an active access policy rule
2. When: The operator attempts to modify the access policy

Expected result: Expected: The system denies the modification request and returns a permission denied error

Evidence to record: Evidence: Test capture of the denial response when attempting to modify policies

Test type: inspection

Id: TEST-009

Requirement ids: SEC-001

Procedure

1. Given: The operator receives an alert for access policy change
2. When: The operator initiates the review action

Expected result: Expected: The system confirms the alert is acknowledged and no access policy modification occurs

Evidence to record: Evidence: Test capture of resulting state shows the alert status updated to ' acknowledged' without any change to access policies, as verified by a real-time policy integrity check

Test type: positive

Id: TEST-010

Requirement ids: SEC-002

Procedure

1. Given: An administrator performs a policy change operation
2. When: The system logs the policy change event

Expected result: Expected: The audit record is stored in the system and remains accessible for 30 days after creation

Evidence to record: Evidence: Test capture of audit records retrieved after 30 days from change timestamp

Test type: positive

Id: TEST-011

Requirement ids: SEC-002

Procedure

1. Given: The system has recorded an administrator policy change
2. When: The system attempts to return the record

Expected result: Expected: The system retains audit records for 30 days

Evidence to record: Evidence: Test capture of resulting state shows audit records retained for 30 days in the log repository with no data loss or corruption over time, as verified by a compliance inspection report from E-005_test_

Test type: analysis

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Access control

Detail: Rules and enforcement governing which principals may perform actions on resources.

Status: proposed

Evidence ids: E-001

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-004; E-005

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-005

Device isolation

Detail: Restriction of a device's network access; the source determines whether and by whom this action is permitted.

Status: proposed

Evidence ids: E-003; E-007

Access restoration

Detail: Returning access that was previously restricted, within the source-defined permission boundary.

Status: proposed

Evidence ids: E-003

Concurrency

Detail: Activity occurring at the same time; it is distinct from the registered planning population.

Status: proposed

Evidence ids: E-006

Registered population

Detail: The inventory of registered users or devices; it does not by itself establish a simultaneous load or a maximum.

Status: proposed

Evidence ids: E-006

Retention

Detail: The period or rule for keeping records. A bare duration does not itself specify deletion timing or minimum/maximum bounds.

Status: proposed

Evidence ids: E-005

Firewall

Detail: A control that permits or restricts network traffic according to rules; mentioning it does not authorize automatic changes.

Status: proposed

Evidence ids: E-007

14. SDLC recommendation and workflow

Selected model: Agile

Justification: The evolving stability (E-008) and frequent feedback (E-008) favor Agile because it enables iterative delivery, continuous adaptation, and rapid response to changing requirements. The absence of budget, failure consequences, or legacy systems makes Hybrid, Incremental, DevSecOps, Spiral, V-Model, and Waterfall less suitable due to insufficient differentiating evidence. No evidence supports budget, delivery, uncertainty, verification, regulation, schedule, or failure consequence thresholds.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: Students and guests may access the internet but must not access the internal staff network.; FR-002: Network administrators may manually isolate a device and restore its access.; FR-003: Do not automatically change firewall rules or isolate devices.
2. Review and baseline the source-defined work group: SEC-001: Security operators may review and acknowledge alerts but may not change access policies.; SEC-002: The system must record administrator policy changes and retain audit records for 30 days.

3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against SEC-001: Security operators may review and acknowledge alerts but may not change access policies.
2. Review implementation and acceptance evidence against SEC-002: The system must record administrator policy changes and retain audit records for 30 days.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: Students and guests may access the internet but must not access the internal staff network.; FR-002: Network administrators may manually isolate a device and restore its access.; FR-003: Do not automatically change firewall rules or isolate devices.
3. Implement the source-defined work group: SEC-001: Security operators may review and acknowledge alerts but may not change access policies.; SEC-002: The system must record administrator policy changes and retain audit records for 30 days.
4. In an authorized isolated test environment, execute TEST-001: Given: A student device connected to the public network When: The student attempts to access the internet Expected: The request is granted and connectivity is established
5. In an authorized isolated test environment, execute TEST-002: Given: A guest device connected to the public network When: The guest attempts to access the internal staff network Expected: The request is denied and no connection is established
6. In an authorized isolated test environment, execute TEST-003: Given: An administrator with valid credentials and a device under management When: The administrator initiates manual isolation of the device via console interface Expected: The device is isolated and status changes to 'isolated' in system state
7. In an authorized isolated test environment, execute TEST-004: Given: An administrator with valid credentials and a device under management When: The administrator initiates manual restoration

of access to the device Expected: Access is restored and device status changes to 'active'

8. In an authorized isolated test environment, execute TEST-005: Given: A test device in active network state with no prior isolation When: The system attempts to automatically isolate the device via scheduled rule update or trigger event Expected: No isolation occurs and the device remains in active state
9. In an authorized isolated test environment, execute TEST-006: Given: A test device with active connection and configured rules When: The system attempts to automatically modify firewall rules to restrict access Expected: No rule modification occurs and the existing rules remain unchanged
10. In an authorized isolated test environment, execute TEST-007: Given: An operator with security privileges and an active alert in the system When: The operator acknowledges the alert Expected: The system returns a successful acknowledgement response indicating the alert is acknowledged
11. In an authorized isolated test environment, execute TEST-008: Given: An operator with security privileges and an active access policy rule When: The operator attempts to modify the access policy Expected: The system denies the modification request and returns a permission denied error
12. In an authorized isolated test environment, execute TEST-009: Given: The operator receives an alert for access policy change When: The operator initiates the review action Expected: The system confirms the alert is acknowledged and no access policy modification occurs
13. In an authorized isolated test environment, execute TEST-010: Given: An administrator performs a policy change operation When: The system logs the policy change event Expected: The audit record is stored in the system and remains accessible for 30 days after creation
14. In an authorized isolated test environment, execute TEST-011: Given: The system has recorded an administrator policy change When: The system attempts to return the record Expected: The system retains audit records for 30 days
15. Collect stakeholder feedback in accordance with the source: Requirements evolve frequently and stakeholders provide weekly stakeholder feedback.
16. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against SEC-001: Security operators may review and acknowledge alerts but may not change access policies.
2. Review implementation and acceptance evidence against SEC-002: The system must record administrator policy changes and retain audit records for 30 days.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against SEC-001: Security operators may review and acknowledge alerts but may not change access policies.
2. Review implementation and acceptance evidence against SEC-002: The system must record administrator policy changes and retain audit records for 30 days.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

Unknown factors: budget; delivery; failure_consequences; legacy; regulation; schedule; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Factors

Name: stability

Value: evolving

Evidence ids: E-008

Name: feedback

Value: frequent

Evidence ids: E-008

Ranked candidates

Model: Agile

Score: 90

Reasons: stability=evolving (E-008); feedback=frequent (E-008)

Model: Hybrid

Score: 72

Reasons: stability=evolving (E-008); feedback=frequent (E-008)

Model: Incremental

Score: 60

Reasons: stability=evolving (E-008)

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 15

Reasons: stability=evolving (E-008)

Evidence register

Id: E-001

Source: description

Quote: Build a campus wireless network access-control requirements specification.

Id: E-002

Source: description

Quote: Students and guests may access the internet but must not access the internal staff network.

Id: E-003

Source: description

Quote: Network administrators may manually isolate a device and restore its access.

Id: E-004

Source: description

Quote: Security operators may review and acknowledge alerts but may not change access policies.

Id: E-005

Source: description

Quote: The system must record administrator policy changes and retain audit records for 30 days.

Id: E-006

Source: description

Quote: The planning population is 2000 registered devices; concurrency is not yet decided.

Id: E-007

Source: description

Quote: Do not automatically change firewall rules or isolate devices.

Id: E-008

Source: description

Quote: Requirements evolve frequently and stakeholders provide weekly stakeholder feedback.

Id: E-009

Source: description

Quote: Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7246

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.6986

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:17 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 17

Page or section: text chunk 18

Relevance score: 0.6875

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:11 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 11

Page or section: text chunk 12

Relevance score: 0.6915

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.6928

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Security, privacy and compliance: qwen3:4b-instruct-2507-q4_K_M

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.28

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-002	UC-001, UC-002, TEST-001, TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-003	UC-003, UC-004, TEST-003, TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-007	TEST-005, TEST-006, Requirements and verification planning, Agile delivery cycle, Release and handoff
SEC-001	E-004	UC-005, UC-006, UC-007, TEST-007, TEST-008, TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff
SEC-002	E-005	TEST-010, TEST-011, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.28